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Optimal Se content on P/G-EGCG@Se nanofibers reverses muscle-derived IL-6- induced osteogenesis depression

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A R T I C L E I N F O

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A B S T R A C T

Objective: To develop electrospun polycaprolactone/gelatin (P/G) nanofibers modified with epigallocatechin gallate-selenium nanoparticles (EGCG@Se) and evaluate their osteogenic effects *in vitro* and *in vivo*.
Methods: P/G nanofibers were fabricated by electrospinning, and EGCG@Se nanoparticles were synthesized using cysteine as a reducing agent. Composite fibers were formed by surface deposition and characterized using SEM, WCA, EDS, FTIR, AFM, and mechanical tests. *In vitro*, MSCs and MC3T3-E1 cells were cultured on P/G-EGCG@Se fibers under osteogenic induction. A tibial defect model in KM mice was used for *in vivo* osteogenic evaluation. Gene and protein expression were analyzed by qPCR, Western blot and ELISA.
Results: P/G fibers with an 8:2 weight ratio and 12 kV voltage exhibited optimal properties. EGCG@Se enhanced MSCs viability, adhesion and osteogenic differentiation. *In vitro*, 7.5 μg/mL Se-loaded fibers showed the best osteoinductive effect. However *in vivo*, 15 μg/mL Se-loaded fibers resulted in the optimal tibial defect repair, which significantly suppressed IL-6 expression in skeletal muscle. *In vitro* co-culturing test demonstrated that Se content in P/G-EGCG@Se nanofibers controls IL-6 production in muscle fibers.
Conclusions: The special physicochemical properties of P/G-EGCG@Se fibers, assisted by bone repair effects of EGCG, and by the optimal Se content induced IL-6 level reduction from muscle tissue, contribute to the efficient bone defect healing *in vivo*.

1. Introduction

An ideal bone repair scaffold should possess both osteoinductive and osteoconductive properties. Among various material systems, electrospun fibers with micro- to nanoscale diameters and porosity offer an effective strategy for large-segment bone defect repair. For example, polylactic acid nanofibers modified with 100–200 nm chitosan “island-like” structures promote osteoblast spreading, mineral deposition and osteogenic activity [1]; aligned electrospun polycaprolactone (PCL) scaffolds enhance osteogenesis of mesenchymal stem cells (MSCs) and accelerate calcium phosphate mineralization [2]; core–shell nanofiber membranes composed of PCL and nano-hydroxyapatite (nHAp) shells with gelatin and metronidazole cores exhibit both osteogenic and antibacterial effects [3]; and silk fibroin/chitosan composite electrospun scaffolds exhibit synergistic effects in bone formation [4]. These studies suggest that by tailoring fabrication processes, optimizing surface nanotopography and fiber alignment, and constructing multiscale fiber architectures, electrospun fibers can mimic the extracellular matrix (ECM) of bone tissue, offering high surface-to-volume ratios that promote cell adhesion, migration, differentiation and polarized growth, thereby facilitating bone regeneration.

Enhancing cell affinity and biological functionality of scaffold surfaces remains a key challenge for improving their osteoinductive and osteoconductive performance. Incorporating bioactive nanoparticles

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into scaffolds is a promising strategy. Previous studies have shown that doping hydroxyapatite with selenium, silicon, or zinc improves its surface bioactivity [5–7]. Nanoparticles such as nano-selenium, gold, and zinc oxide have demonstrated osteoregenerative potential and can promote osteogenic differentiation of stem cells [5–7]. Among them, selenium (Se) is an essential trace element that regulates numerous physiological functions, including antioxidant defense, anti-inflammation, bone metabolism and immune modulation [8]. Nano-selenium (Se NPs), with a high surface area-to-volume ratio, can form covalent or non-covalent interactions with various biomolecules, metabolites, and immune cells, and serves as a stable carrier for proteins, peptides, nucleic acids and drugs. Its applications in disease therapy and tissue repair have drawn increasing attention [9–17]. Recent studies suggest that Se NP-based composites can enhance stem cell osteogenesis and serve as candidate materials for bone scaffolds and orthopedic implants [18,19].

Epigallocatechin gallate (EGCG) has been shown to promote osteoblast differentiation, upregulate osteogenic gene expression, increase alkaline phosphatase (ALP) activity, suppress osteoclastogenesis, and enhance mineralized nodule formation [20–23]. In this study, EGCG was conjugated to selenium nanoparticles using cysteine as a reducing agent to synthesize EGCG@Se. These nanoparticles were subsequently deposited onto the surface of electrospun polycaprolactone/gelatin (PCL/Gelatin) nanofibers *via* surface deposition to form PCL/Gelatin-EGCG@Se (P/G-EGCG@Se) composites. *In vitro* studies demonstrated that the surface topography of the composite fibers modulated MSCs adhesion and viability, and that a lower Se-loaded nanofibers (7.5 μg/mL) favored MSCs and MC3T3-E1 cells proliferation and osteogenic differentiation. Based on these findings, we performed *in vivo* implantation experiments using P/G-EGCG@Se to repair bone defects and identified that the fibers loading with 15 μg/mL Se showed optimal bone regeneration. Our further investigation revealed that the Se content-dependent suppression of IL-6 synthesis from myocytes played an important regulatory role for osteogenesis (Scheme 1). This study provides a rationale for the design of novel biomaterials that guide bone regeneration through interactions with the surrounding non-bone tissues.

2. Materials and methods

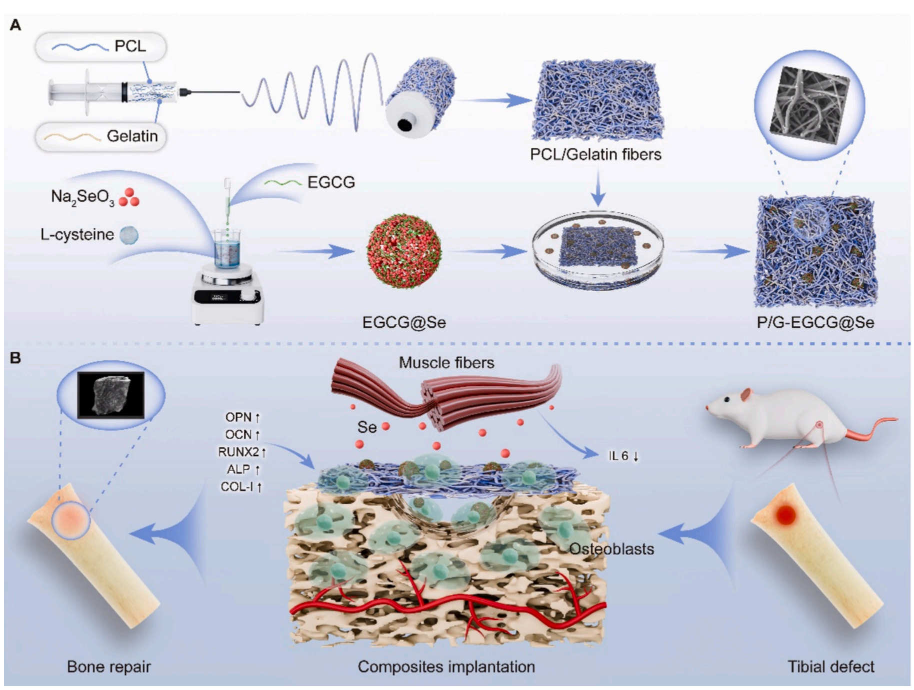
2.1. Ethical approval

All animal procedures were approved by the Institutional Animal Care and Use Committee of Southern Medical University (Approval No. L2016068). Kunming (KM) mice were purchased from the Laboratory Animal Center of Southern Medical University (Guangzhou, China). Animals were housed in a controlled environment (23 °C, 50% relative humidity, 12-h light/dark cycle) with *ad libitum* access to standard chow and water.

2.2. Synthesis of P/G-EGCG@Se nanofibers

Electrospinning solutions were prepared using polycaprolactone (PCL, molecular weight ~ 80,000; Daigang Bioengineering Co., Jinan , China.) and type A gelatin (Sigma-Aldrich) at a mass ratio of 8:2. Specifically, 10 wt% PCL and 4 wt% gelatin were dissolved separately in hexafluoroisopropanol (HFIP) and stirred magnetically for 24 h before mixing. Electrospinning was performed using a high-voltage power supply (LDHV, Dongwen High Voltage Co., China) and syringe pump (TYD01, Lead Fluid, China). The mixed solution was loaded into a 10 mL syringe and delivered through a 20-gauge blunt-tip needle at a flow rate of 2 mL/h. A voltage of ~18 kV was applied between the needle tip and a grounded aluminum collector placed 12–15 cm away. Electrospinning was conducted at room temperature under ventilated conditions. The resulting randomly oriented fiber membranes were dried under ambient conditions to remove residual solvent. For aligned fibers, a rotating drum collector (89 mm diameter) was used, with the needle positioned ~10 cm away. The drum was rotated at 2500 rpm, and the voltage was gradually adjusted between 10 and 18 kV to align fibers along the rotation direction.

EGCG@Se nanoparticles were synthesized using cysteine as a reducing agent. Sodium selenite (Na_2SeO_3) and L-cysteine were mixed, followed by dropwise addition of epigallocatechin gallate (EGCG, 0.05 mol/L) under constant stirring. The resulting EGCG-functionalized selenium nanoparticles (EGCG@Se) were purified by centrifugation and



Scheme 1. The fabrication process of P/G-EGCG@Se nanofibers (A) and which enables the modulation of the muscle-derived IL-6-impaired bone formation *in vivo* (B).

redispersed in water. Cysteine@Se particles (without EGCG) were prepared under identical conditions as controls.

Surface deposition was used to load selenium nanoparticles onto fiber membranes. P/G fibers (8:2 mass ratio) were immersed in EGCG@Se suspensions containing elemental selenium at 3.75, 7.5, 15, 30, or 60 μg/mL for 12 h with gentle shaking at room temperature. After deposition, membranes were washed with deionized water and air-dried. Control P/G membranes (without EGCG@Se) underwent the same procedure. Additional control fibers were prepared using Cysteine@Se suspensions of matching Se concentrations.

2.3. Characterization of P/G-EGCG@Se nanofibers

Fiber morphology and diameter were evaluated using scanning electron microscopy (SEM, JSM-1400F, JEOL, Japan) at 10 kV. Samples were sputter-coated with gold for 60 s prior to imaging. Hydrophilicity was assessed *via* static water contact angle measurement (OCS15EC, Dataphysics, Germany) by applying 10 μL distilled water and recording the angle at 2 s. Mechanical properties (tensile strength, Young's modulus, and elongation at break) were measured using a universal testing machine. Degradation tests were conducted by immersing samples in PBS (37 °C) and weighing them at defined intervals to calculate mass loss. Fiber alignment was analyzed from SEM images using directionality mapping.

The morphology of EGCG@Se and Cysteine@Se nanoparticles was characterized by transmission electron microscopy (TEM, H-7650, Hitachi, Japan). Hydrodynamic size and zeta potential were determined *via* dynamic light scattering (DLS, Malvern Zetasizer). EGCG surface modification was confirmed by ATR-FTIR (Nicolet iS50, Thermo Fisher) and UV–Vis spectroscopy, with a characteristic EGCG peak observed at 275 nm.

Atomic force microscopy (AFM, Cypher VRS, Oxford Instruments) was used to examine fiber surface topography over 5 μm × 5 μm scan areas and quantify roughness. Elemental mapping by energy-dispersive spectroscopy (EDS) assessed selenium distribution. FTIR spectra were used to evaluate chemical changes after nanoparticle deposition.

The Se loading efficiency in composite fibers was assessed through inductively coupled plasma mass spectrometry (ICP-MS). Specifically, P/G-EGCG@Se nanofiber, fabricated with varying concentrations of immersion solutions, were sectioned into specimens of comparable size ($n = 3$). The Se content within the membranes was quantified using ICP-MS. Subsequently, the Se loading efficiency (LE, %) was calculated according to the following equation (the ratio of the actual Se mass loaded on the scaffold to the total Se mass in the initial immersion suspension):

$$\mathrm{LE} = \frac{W_{\mathrm{actual}}}{W_{\mathrm{initial}}} \times 100\%$$

To evaluate the release kinetics of Se from the composite fibers, the *in vitro* cumulative release profile was determined using Inductively Coupled Plasma Mass Spectrometry (ICP-MS). Briefly, P/G-EGCG@Se nanofibers were cut into specimens of uniform weight ($n = 3$). The samples were immersed in glass vials containing 5 mL of PBS (pH 7.4) and incubated at 37 °C under constant shaking (100 rpm). At predetermined time points (1, 3, 5, 7, and 14 days), 1 mL of the supernatant was withdrawn for analysis and immediately replaced with an equal volume of fresh, pre-warmed PBS to maintain constant volume. The concentration of released Se in the collected samples was quantified by ICP-MS. The cumulative release percentage was calculated using the following formula:

$$\text{Cumulative Release (\%)} = \frac{\text{Total amount of Se released at time t}}{\text{Total initial Se loading amount}} \times 100\%$$

The detailed microstructure and elemental distribution of the nanoparticles were characterized using a High-Resolution Transmission Electron Microscope (HRTEM, Talos F200X G2, Thermo Fisher Scientific) operated at an accelerating voltage of 200 kV. For sample preparation, a drop of the nanoparticle suspension was placed onto a carbon-coated copper grid and air-dried at room temperature. The morphology was observed in High-Angle Annular Dark-Field (HAADF) mode. Elemental composition and distribution were analyzed using Energy Dispersive X-ray Spectroscopy (EDS) mapping. The crystalline phase of the nanoparticles was determined by Selected Area Electron Diffraction (SAED).

2.4. Co-culture of nanofibers with cells

Mesenchymal stem cells (MSCs) or MC3T3-E1 pre-osteoblasts were cultured in complete medium containing 10% fetal bovine serum (FBS, Gibco) until reaching 80% confluency. The cells were then seeded into culture systems containing P/G-EGCG@Se nanofibers containing different Se content or floating P/G fibers, and subjected to osteogenic induction. The osteogenic medium consisted of 0.2 mM ascorbic acid, 10 mM β-glycerophosphate and 10^{-8} M dexamethasone. The medium was refreshed every 3 days.

To investigate the effect of muscle-derived cytokines on bone formation, C2C12 myoblasts were firstly differentiated for 72 h in medium containing 2% horse serum (Gibco, USA). MC3T3-E1 cells were seeded in the upper chambers of Transwell inserts, and osteogenic induction medium was added once the cells reached 80% confluency. Differentiated C2C12 myotubes were placed in the lower chambers, and co-cultured with P/G-EGCG@Se nanofibers containing different Se content or with P/G fibers. To elevate IL-6 level within the co-culture system, recombinant mouse interleukin-6 (rmIL-6, 10 ng/mL, Abbkine) was added. For blocking IL-6 signaling, IL-6 neutralizing antibody (10 ng/mL, Abcam) or gp130 antaogonist SC144 (1 μM, MedChemExpress) was further added into co-culture respectively. The medium was refreshed every 3 days.

2.5. In vivo evaluation of nanofibers

All animal experiments adhered to ethical guidelines. Adult male KM mice were anesthetized, and a longitudinal incision (~6 mm) was made on the anterior tibia near the tibial tuberosity. The tibialis anterior (TA) muscle was bluntly dissected, and a 2 mm K-wire was inserted to induce muscle injury. A 0.8 mm cortical defect was created using a K-wire drill. The defect was filled with P/G or P/G-EGCG@Se fibers containing different Se content (1 mg/per defect). Sham-operated mice received no implants. Skin was sutured, and penicillin (2×10^4 U) was administered subcutaneously. Mice were housed individually with free access to food and water.

For elevating plasma IL-6 level of model mice, recombinant mouse IL-6 (rmIL-6, 100 ng, Abbkine) were injected into tail vein of model mice on day 1 after implanting with P/G-EGCG@Se fibers [24].

2.6. PCR analysis

Total RNA was extracted from MSCs, MC3T3-E1 cells, C2C12 cells, tibial defects, or injured TA muscle using Trizol (Invitrogen, USA). RNA purity and concentration were measured by UV spectrophotometry. 1 mg of RNA was reverse-transcribed into cDNA using the PrimeScript RT kit (Takara, Japan). qPCR was performed with SYBR Green reagents (Fermentas) using an ABI StepOne Plus system (Applied Biosystems). Each 25 μL reaction included 2 μL cDNA, 0.5 μL of each primer (10 μM), 12.5 μL SYBR mix, and nuclease-free water. Cycling parameters: 95 °C for 30 s, then 40 cycles of 95 °C for 5 s and 60 °C for 30 s. Melt curve analysis ensured specificity. Gene expression was normalized to GAPDH using the 2^-ΔΔCt method. Primer sequences were listed in Table 1 (synthesized by Sangon Biotech, Shanghai, China).

Table 1
Gene primer sequences.

Genes	Primer sequences
Runx2	Forward 5′- TGGTTACTGTCATGGCGGGTA −3′ Reverse 5′- TCTCAGATCGTTGAACCTTGCTA −3′
OPN	Forward 5′ - CTCCATTGACTCGAACGACTC-3′ Reverse 5′ - CAGGTCTGCGAAACTTCTTAGAT −3′
OCN	Forward 5′ - CACTCCTCGCCCTATTGGC-3′ Reverse 5′ - CCCTCCTGCTTGGACACAAAG −3′
ALP	Forward 5′ -AACATCAGGGACATTGACGTG-3′ Reverse 5′ - GTATCTCGGTTTGAAGCTCTTCC -3′
IL-6	Forward 5′ - GAGTCCGCAGCAGGTG-3′ Reverse 5′ - GTGTCAGAGTCCATGGGA-3′
GAPDH	Forward 5′ -TATGACTCTACCCACGGCAAG-3′ Reverse 5′ -ATACTCAGCACCAGCATCACC-3′

2.7. Western blot analysis

Cells or tissues were lysed in ice-cold RIPA buffer supplemented with protease inhibitor cocktail for 10 min. Lysates were centrifuged at 12,000 rpm for 10 min, and protein concentrations were quantified using a BCA protein assay. Equal amounts of protein (30 μg per sample) were mixed with 5× SDS loading buffer, denatured at 95 °C for 5 min, and separated by SDS-PAGE. Proteins were then transferred onto 0.22 μm PVDF membranes and blocked with 5% non-fat milk in TBST for 1 h at room temperature. Membranes were incubated overnight at 4 °C with primary antibodies: rabbit anti-mouse Desmin (CST, Boston, MA, 1:1000), rabbit anti-mouse Myogenin (Invitrogen, Carlsbad, CA, 1:1000), rabbit anti-mouse Runx2 (Cell Signaling Technology, 1:1000), rabbit anti-mouse OPN (Abcam, 1:1000), rabbit anti-mouse gp130 (Wanleibio, China, 1:1000), rabbit anti-mouse ALP (Abcam, 1:1000), rabbit anti-mouse OCN (Abcam, 1:1000), rabbit anti-mouse COL-I (Abcam, 1:1000), rabbit anti-rabbit GAPDH (Abcam, 1:1000), or rabbit anti-mouse β-Actin (Sigma, 1:1000). After washing 3 times with TBST (5 min each), membranes were incubated with HRP-conjugated secondary antibodies (anti-mouse or anti-rabbit IgG, Fudebio, 1:5000) for 1 h at room temperature. Bands were visualized using enhanced chemiluminescence (ECL) reagents, and images were captured using a gel imaging system (Protein Simple, CA). Band intensities were quantified using ImageJ (v1.42, NIH, Bethesda, MD) and normalized to β-Actin or GAPDH.

2.8. Histological and immunofluorescence analysis

Bone tissues from the defect area were fixed in 10% neutral-buffered formalin for 48 h and decalcified in 10% EDTA for 4 weeks, followed by paraffin embedding and sectioning at 5 μm. Sections were stained with H&E or subjected to immunofluorescence using the following protocol: antigen retrieval with sodium citrate buffer (pH 6.0) at 125 °C for 2 min, blocking, and incubating with rabbit anti-mouse gp130 (Wanleibio, China, 1:1000) at 4 °C overnight. Fluorescent secondary antibody (555-donkey anti-rabbit IgG, Beyotime, China, 1:1000) and DAPI counterstaining (Abcam, Eugene, OR, USA) were applied.

Frozen sections of tibialis anterior (TA) muscle (8 μm) were fixed in ice-cold acetone, blocked, and incubated with rabbit anti-mouse Desmin (CST, 1:500), rabbit anti-mouse Myogenin (Invitrogen, 1:500), or mouse anti-mouse Dystrophin (mouse anti-mouse, Proteintech, 1:500) overnight at 4 °C. Secondary antibodies included 555-donkey anti-rabbit IgG or 488-goat anti-rabbit IgG (Beyotime, China, 1:1000). Nuclei were counterstained with DAPI. Images were acquired with a fluorescence microscope (Olympus BX51).

For MSCs or MC3T3-E1 cells, fixation was performed with 4% paraformaldehyde for 15 min, followed by PBS washing and membrane permeabilization with 0.1% Triton X-100 for 10 min. After blocking with 2% BSA for 1 h, cells were incubated overnight at 4 °C with rabbit anti-mouse Vinculin (Servicebio, 1:1000) and mouse anti-mouse Actin (Phalloidin, Servicebio, 1:1000). After PBS washes, fluorescent secondary antibodies were added and incubated at room temperature for 1 h. DAPI counterstaining was performed prior to imaging.

2.9. Alizarin Red, ALP, and live/dead staining

MSCs were washed with PBS, fixed in 4% paraformaldehyde for 15 min, and stained with 40 mM Alizarin Red S solution (pH 4.2, in deionized water; Beyotime, China) for 15 min at room temperature. Samples were washed with distilled water until no red dye remained in the wash solution. Mineralized nodules were observed under an inverted microscope.

ALP activity in MC3T3-E1 cells was detected using a modified Gomori calcium-cobalt staining kit (Solarbio, G1481, China). For viability assays, MC3T3-E1 cells were incubated with Calcein-AM/PI working solution (2 μM Calcein-AM +4 μM PI in PBS) for 15 min at 37 °C in the dark. Fluorescent images were obtained under an inverted fluorescence microscope, and green (live) and red (dead) cells were counted in randomly selected fields.

2.10. Micro-CT analysis

Bone specimens harvested at 2 and 4 weeks post-operation were scanned using a Micro-CT scanner (Bruker SkyScan 1176). Scanning parameters: 90 kV voltage, 80 μA current, 9 μm resolution, 0.5° rotation step, and 180° scanning range. Three-dimensional reconstruction was performed using the built-in software. Bone regeneration was evaluated based on bone volume/total volume (BV/TV), trabecular number, and 3D visualization of reconstructed images.

2.11. Proteomics data analysis

Publicly available dataset GSE594 from the GEO database, representing murine bone tissue proteomes at various fracture time points, was used. Data were imported *via* GEOquery, and analyzed using the limma package in R. Differentially expressed proteins were identified using thresholds of $|\log_2FC| \geq 1$ and FDR < 0.05. Protein-protein interaction networks, GO enrichment, heatmaps, and volcano plots were generated to investigate dynamic changes and functional roles of key molecules during bone repair. Visualization was performed using ggplot2 and pheatmap. Specific analysis results are not detailed in this section.

2.12. Statistical analysis

All data were obtained from at least 3 independent experiments and expressed as mean ± standard deviation (SD). Statistical analyses were conducted using GraphPad Prism 8.0. Two-group comparisons were made using unpaired Student's *t*-test, while one-way ANOVA followed by Tukey's *post hoc* test was used for multiple group comparisons. Statistical significance was set as follows: $*p < 0.05$, $**p < 0.01$, $***p < 0.001$, $****p < 0.0001$.

3. Results

3.1. Characterization of the P/G-EGCG@Se electrospun nanofiber composites

Polycaprolactone/Gelatin (PCL/Gelatin, P/G) nanofibers were produced through the electrospinning technique. Scanning electron microscopy (SEM) analysis indicated that fibers with a P/G mass ratio of 0:10 exhibited significant aggregation. Conversely, fibers with P/G ratios of 10:0, 8:2, 5:5, and 2:8 demonstrated average diameters of 632 ± 196 nm, 386 ± 61 nm, 374 ± 52 nm, and 274 ± 37 nm, respectively (Supplementary Fig. 1). Water contact angle (WCA) measurement revealed the values of 112.9 ± 3.6°, 63.4 ± 2.2°, 48.9 ± 1.9°, 38.0 ± 3.0°, and 33.7 ± 3.7° separately (Supplementary Fig. 2), indicating that an increase in gelatin content corresponded with enhanced fiber hydrophilicity. Analysis of stress-strain curves and Young's modulus demonstrated that fibers with a P/G ratio of 8:2 exhibited the highest tensile strength and elongation at break (Supplementary Fig. 3). Immersion tests further validated that the 8:2 P/G fibers experienced minimal mass loss and exhibited superior stability (less than 20%) (Supplementary Fig. 4), along with the slowest degradation rate (Supplementary Fig. 5). SEM imaging also illustrated that fibers electrospun at 12 kV with a PCL/Gelatin ratio of 8:2 displayed the most uniform orientation (Supplementary Fig. 6).

Subsequently, we synthesized EGCG@Se nanoparticles by conjugating epigallocatechin gallate (EGCG) with selenium nanoparticles utilizing cysteine as a reducing agent. Transmission electron microscopy (TEM) revealed that both EGCG@Se and Cysteine@Se nanoparticles exhibited a spherical morphology (Fig. 1A, B), with average diameters of 221 ± 3 nm and 186 ± 2 nm, respectively (Fig. 1C). Their Zeta potentials were measured at −28.8 mV and − 25.1 mV, respectively (Fig. 1D). The Fourier-transform infrared (FTIR) spectrum of EGCG@Se displayed characteristic peaks at 3356–3550 cm^{-1}, 1223–1237 cm^{-1}, and 1040–1147 cm^{-1}, which correspond to the stretching vibrations of -OH, -O=C-O, and -C-O groups attributed to EGCG (Fig. 1E). The UV–Vis absorption spectrum exhibited a characteristic peak at 275 nm for both EGCG and EGCG@Se, thereby confirming the successful modification of selenium nanoparticles with EGCG (Fig. 1F).

As illustrated in Fig. 1G and H, HAADF-STEM image revealed that EGCG@Se nanoparticles possessed a uniform spherical morphology.

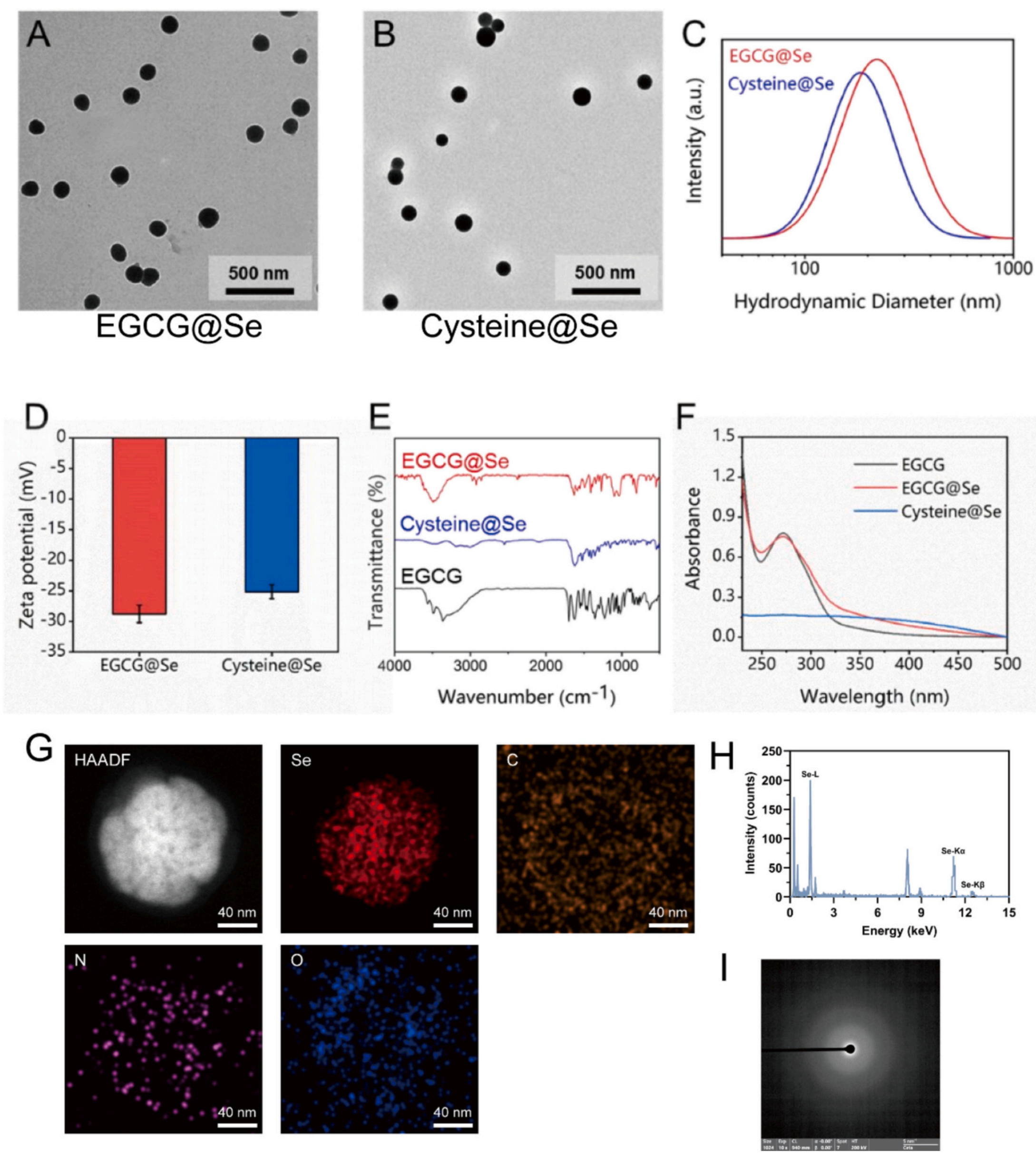


Fig. 1. Physicochemical charaterization of EGCG@Se and Cysteine@Se nanoparticles.
Transmission electron microscopy (TEM) images of EGCG@Se (A) and Cysteine@Se (B). Particle size distribution (C), Zeta potential measurement (D), Fourier-transform infrared (FTIR) spectra (E) and UV–visible absorption spectra (UV–Vis) analysis (F) confirming the successful coordination of EGCG or L-cysteine with Se. High-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) image of EGCG@Se nanoparticles showing a uniform spherical morphology (G), together with corresponding elemental mapping images of Se, C, N, and O, demonstrating a homogeneous elemental distribution throughout the nanoparticles (H). Selected area electron diffraction (SAED) pattern of EGCG@Se nanoparticles exhibiting a diffuse halo ring without distinct diffraction spots, indicative of an amorphous Se structure (I). Statistical analysis was performed using one-way ANOVA followed by Tukey's *post hoc* test. All data are presented as mean ± standard deviation (n = 3, independent experiments). Bar = 500 nm.

Elemental mapping demonstrated a homogeneous distribution of Se, which was colocalized with carbon (C), nitrogen (N), and oxygen (O) throughout the nanoparticle. The signals corresponding to carbon, nitrogen and oxygen are attributed to the presence of EGCG and L-cysteine. Additionally, the Selected Area Electron Diffraction (SAED) pattern displayed a diffuse halo ring lacking distinct diffraction spots, suggesting that the Se core exists in an amorphous rather than crystalline phase (Fig. 1I). This amorphous structure is commonly associated with enhanced bioavailability and is characteristic of Se nanoparticles stabilized by proteins or polysaccharides.

P/G nanofiber membranes, formulated with a mass ratio of 8:2, were subjected to immersion in equal volumes of EGCG@Se or Cysteine@Se solutions at various selenium concentrations (60, 30, 15, 7.5 and 3.75 μg/mL) for a duration of 12 h to produce the corresponding composite fiber samples. Elemental mapping analysis indicated a pronounced increase in selenium content on the composite fibers correlating with the elevated concentrations of Se nanoparticles in the solution (Fig. 2A). Atomic force microscopy (AFM) assessments revealed that the surface roughness of the fibers escalated with increasing selenium concentration, with the most significant roughness recorded in 60 μg/mL Se-loaded P/G-EGCG@Se fibers (Fig. 2B, C). FT-IR spectroscopy did not exhibit any new peaks before and after the deposition of nanoparticles, suggesting that no new chemical groups were formed (Fig. 2D). Water contact angle measurements for pure P/G fibers, P/G-EGCG@Se at 60 μg/mL and P/G-Cysteine@Se fibers were recorded at 61.9 ± 3.5°, 45.2 ± 2.2°, and 44.3 ± 1.8°, respectively. No statistically significant differences in contact angle were observed among the P/G-EGCG@Se samples at 3.75, 7.5, 15, or 30 μg/mL Se (Fig. 2E). These findings indicate that the deposition of EGCG@Se nanoparticles onto the surface of P/G fibers effectively reduced the contact angle, thereby enhancing the hydrophilicity of the fibers. By analysis of stress-strain curves and Young's modulus, we did not monitor the significant differences in tensile strength and elongation at break among P/G-EGCG@Se fibers at 3.75, 15 or 60 μg/mL Se (Fig. 2F, G, H, I). By ICP-MS, Se content and Se loading efficiency of the fibers were quantified and presented in Table 2. We further characterized the *in vitro* release profile of Se from 15 μg/mL Se-loaded fibers. As shown in Fig. 2J, the fiber demonstrated a cumulative release of ~28% on day 1 and reached ~72% by day 14.

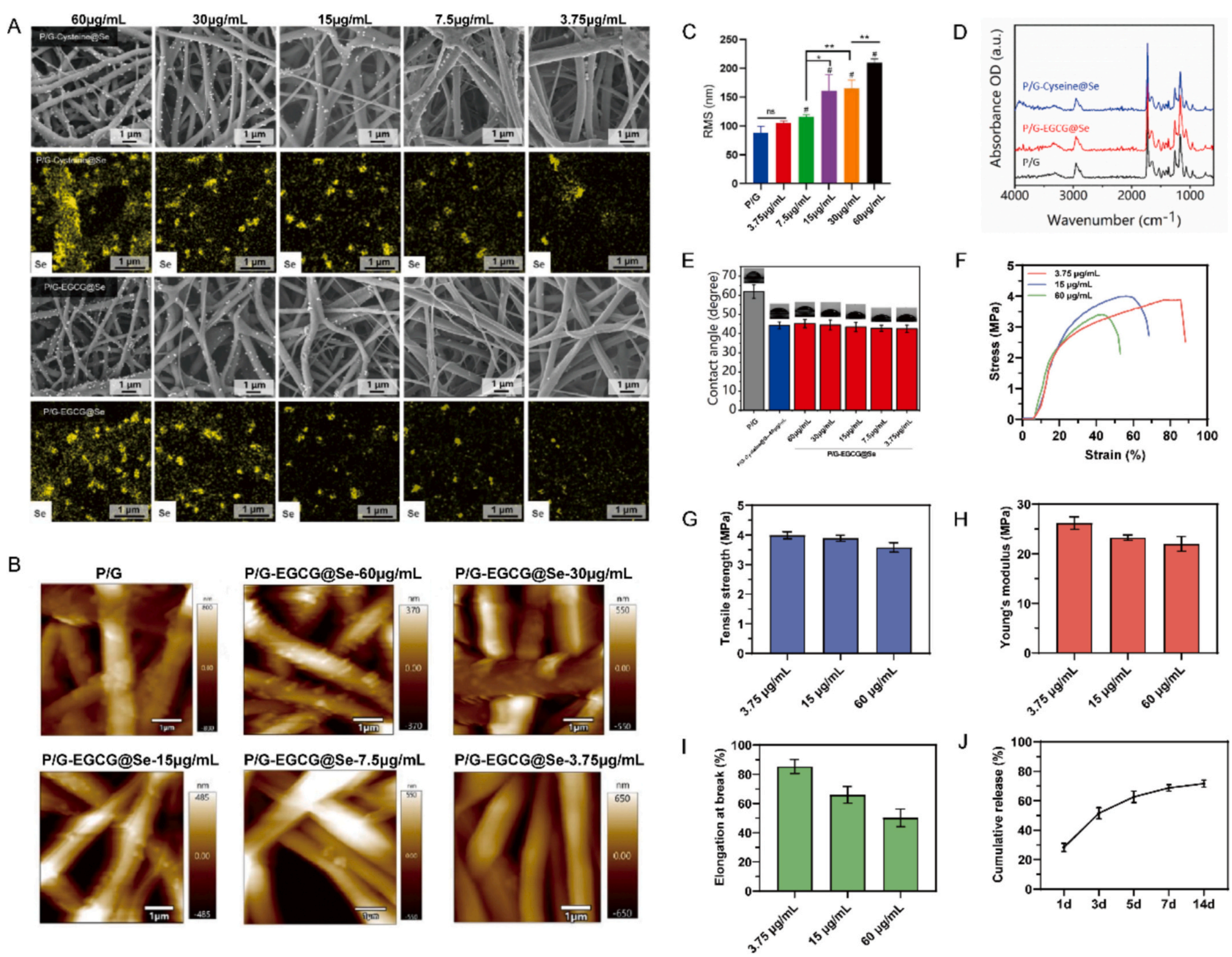


Fig. 2. Characterization of P/G composite nanofibers after deposition of EGCG@Se or Cysteine@Se at different concentrations.
A: Scanning electron microscopy (SEM) images and energy-dispersive X-ray spectroscopy (EDS) mapping of selenium (Se) element distribution in composite nanofibers. B: Atomic force microscopy (AFM) topographic images of fibers. C: Surface roughness analysis of fibers. D: Fourier-transform infrared (FTIR) spectra of fibers before and after nanoparticle deposition. E: Water contact angle measurement of composite nanofibers. F–I: Stress–strain curves and corresponding Young's modulus, tensile strength, and elongation at break of P/G-EGCG@Se fibers loaded with 3.75, 15, and 60 μg/mL Se. J: *In vitro* cumulative selenium release profile of P/G-EGCG@Se fibers loaded with 15 μg/mL Se. Statistical analysis was performed using one-way ANOVA followed by Tukey's *post hoc* test. Statistical significance is indicated as follows: *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$, ****$p < 0.0001$. All data are presented as mean ± standard deviation ($n = 3$, independent experiments). Bar = 1 μm.

Table 2
Se content and Se loading efficiency of the fibers.

Immersion Se Concentration (μg/mL)	Se Loading (μg)	Loading Efficiency (%)
3.75	0.21 ± 0.03	56.0 ± 7.2
7.5	0.52 ± 0.04	69.3 ± 5.1
15	0.87 ± 0.05	58.0 ± 3.2
30	1.15 ± 0.08	38.3 ± 2.7
60	1.42 ± 0.11	23.7 ± 1.8

3.2. P/G-EGCG@Se nanofibers induce osteogenic differentiation of MSCs and MC3T3-E1 cells in vitro

To assess the cytocompatibility, proliferative capacity, and osteoinductive potential of the P/G-EGCG@Se composite nanofibers, we performed co-culture experiments utilizing bone marrow-derived mesenchymal stem cells (MSCs) firstly. The findings indicated that the inclusion of EGCG significantly improved MSCs viability (Supplementary Fig. 7), adhesion (Supplementary Fig. 8), and osteogenic differentiation on the fiber surface, presented by the upregulation of gene and protein expression of osteogenic markers, including Runx2, OPN and OCN in MSCs (Supplementary Fig. 9). Furthermore, the integration of Se nanoparticles exhibited a synergistic effect with EGCG, leading to the increase of calcium deposition (Supplementary Fig. 10). It was noted that the content of Se within the composite fibers had a notable impact on MSCs activity and functionality. Nanofibers with less Se content were found to be more helpful for MSCs proliferation and osteogenic differentiation, especially for nanofibers loaded with 7.5 μg/mL Se, which were identified as the optimal scaffold for facilitating osteogenesis (Supplementary Fig. 9 and Fig. 10).

In agreement with the observations from MSCs, co-culture experiments involving MC3T3-E1 preosteoblasts and the composite fibers also revealed the highest cellular activity at 7.5 μg/mL Se content (Fig. 3A). Following a 14-day period of osteogenic induction, Western blot analysis demonstrated that 7.5 μg/mL Se-loaded P/G-EGCG@Se fibers significantly upregulated osteogenic protein expression in MC3T3-E1 cells (Fig. 3B), which was further supported by alkaline phosphatase

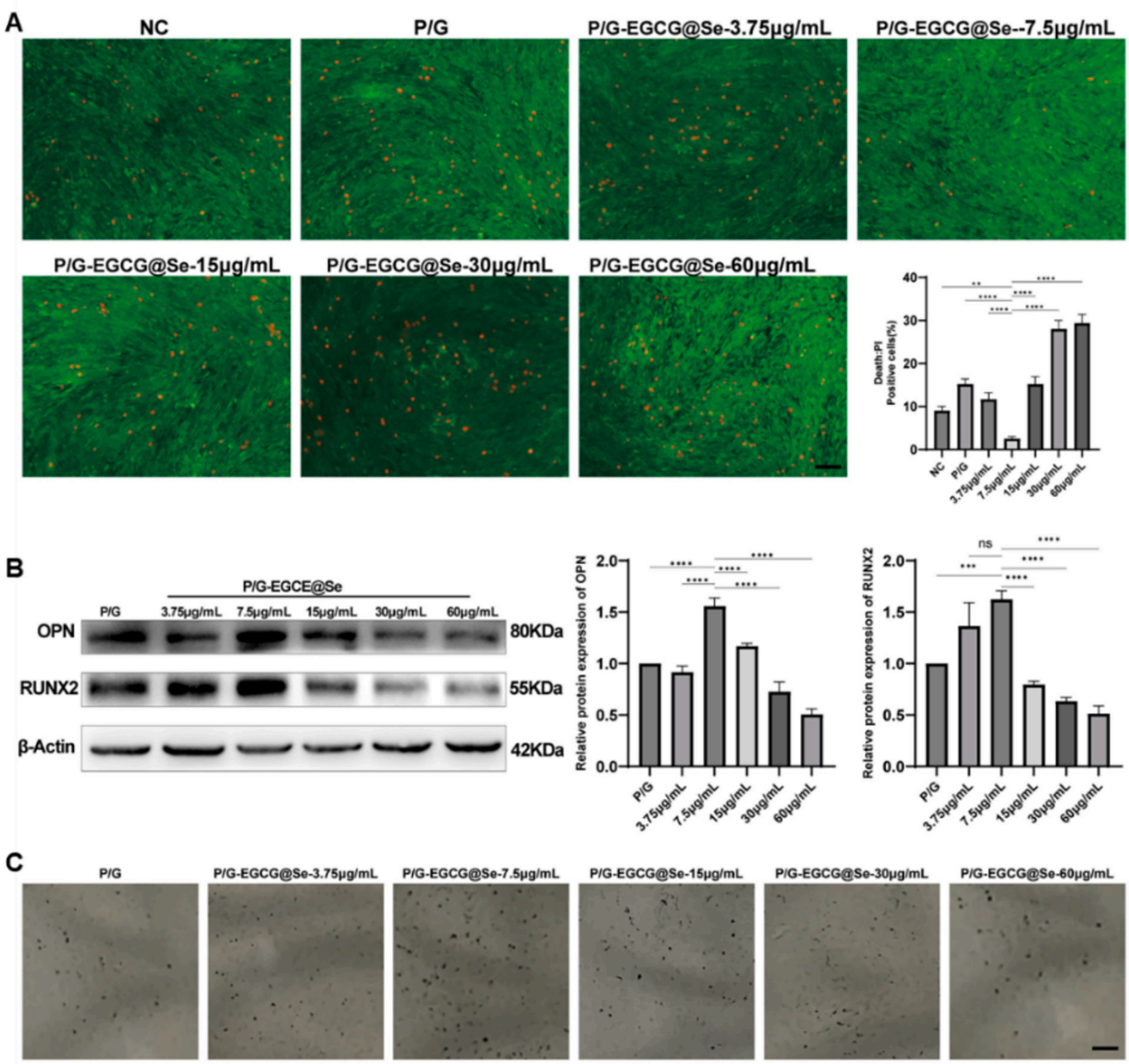


Fig. 3. Osteogenic effect analysis of P/G-EGCG@Se composite nanofibers on MC3T3-E1 cells.
A: Cell viability was assessed by Calcein-AM/PI double staining (Bar = 50 μm). B: Western blot analysis of osteogenesis-related protein expression. C: Alkaline phosphatase (ALP) staining (Bar = 100 μm). Statistical analysis was performed using one-way ANOVA followed by Tukey's *post hoc* test. Statistical significance is indicated as follows: *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$, ****$p < 0.0001$. All data are presented as mean ± standard deviation ($n = 3$, independent experiments).

(ALP) staining (Fig. 3C).

3.3. Analysis of osteogenic effect in vivo induced by P/G-EGCG@Se nanofibers

How is the *in vivo* osteogenic effect of P/G-EGCG@Se composites and does it varies due to the different Se content on composites? To explore this question, we prepared Kunming (KM) mice model of unicortical drill-hole defect in tibia (0.8 mm in diameter) [25–27]. P/G-EGCG@Se nanofibers loaded with 3.75, 7.5, 15, 30, or 60 μg/mL Se, were implanted respectively (Fig. 4A). Unlike the results obtained *in vitro*, our study revealed that nanofibers loaded with 15 μg/mL Se, but not 7.5 μg/mL Se, facilitated the most prominent bone regeneration. Micro-CT imaging conducted four weeks post-implantation indicated nearly complete healing of the bone defect in mice treating with 15 μg/mL Se-loaded nanofibers (Fig. 4B). Histological analysis on two and four weeks post-implants also demonstrated that bone callus formation was most prominent in mice received nanofibers at 15 μg/mL Se comparing to the others (Fig. 4C). Using immunoblotting analyses for Runx2 and OPN proteins in regenerated bone tissue, we further proved that 15 μg/mL Se-loaded nanofibers was the optimal selection for osteogenesis *in vivo* (Fig. 4D). When we focused on inflammation response, we observed F4/80^{+} macrophages invasion in osteotylus tissue. Interestingly, the lower concentration of Se loaded-nanofibers provoked a fewer macrophages infiltration than nanofibers at higher Se. Noteworthy, 15 μg/mL Se-loaded fibers were favorable for controlling inflammation development in osteotylus tissue (Fig. 4E). Together, our work suggests that bone repair *in vivo* is more complex, and it is possible that, Se content on nanoparticles may affect the functionality of other tissues and cells besides regulating bone regeneration. The interaction between bone and non-bone tissues contributes to the osteogenesis and bone healing outcomes.

3.4. Se content in P/G-EGCG@Se nanofibers controls muscle-derived IL-6-depressed bone formation

The P/G-EGCG@Se nanofiber treated with 7.5 μg/mL of Se exhibited the highest *in vitro* osteoinductive potential; however, *in vivo* studies indicated that the scaffold loaded with 15 μg/mL of Se was more effective in facilitating bone defect repair. Skeletal muscle, which is closely associated with bone tissue, secretes various myokines that play a significant role in bone growth, development, repair, and remodeling following injury [28,29]. We posited that selenium element within the composite may influence the secretory activity of muscle fibers after nanofibers implantation, thereby affecting the bone healing process. To investigate this hypothesis, we firstly analyzed a proteomics dataset related to bone repair (GSE594). Network and enrichment analyses indicated that interleukin-6 (IL-6), a prominent myokine produced by skeletal muscle, is significantly involved in the process of bone regeneration (Supplementary Fig. 11A, B). Notably, plasma levels of IL-6 exhibited a marked increase three days post-fracture, followed by a rapid decrease, ultimately returning to near-baseline levels by day seven (Supplementary Fig. 11C, D).

Considering that IL-6 functions as a pleiotropic cytokine [30], we conducted a further investigation into its role in the bone repair process induced by P/G-EGCG@Se nanofibers. The results obtained from enzyme-linked immunosorbent assay (ELISA) indicated that mice implanted with P/G-EGCG@Se nanofibers exhibited significantly reduced plasma IL-6 levels at days 7, 14, and 28, in comparing to P/G nanofiber. Notably, animals received the nanofiber treating with 15 μg/mL of EGCG@Se demonstrated the lowest IL-6 levels (Fig. 5A). Consistently, our western blotting and immunofluorescence analyses further proved that, gp130, a component of IL-6 receptor, significantly down-regulated in osteotylus tissue after implanting of nanofiber treated with 15 μg/mL of EGCG@Se (Fig. 5B, C). For further clarifying the role of IL-6 signaling on bone regeneration in mice received 15 μg/mL Se loaded-fibers, we re-elevate plasma IL-6 level of model mice by injected *i.v.* rmIL-6. As expectedly, the data of Micro-CT imaging (Fig. 5D), HE

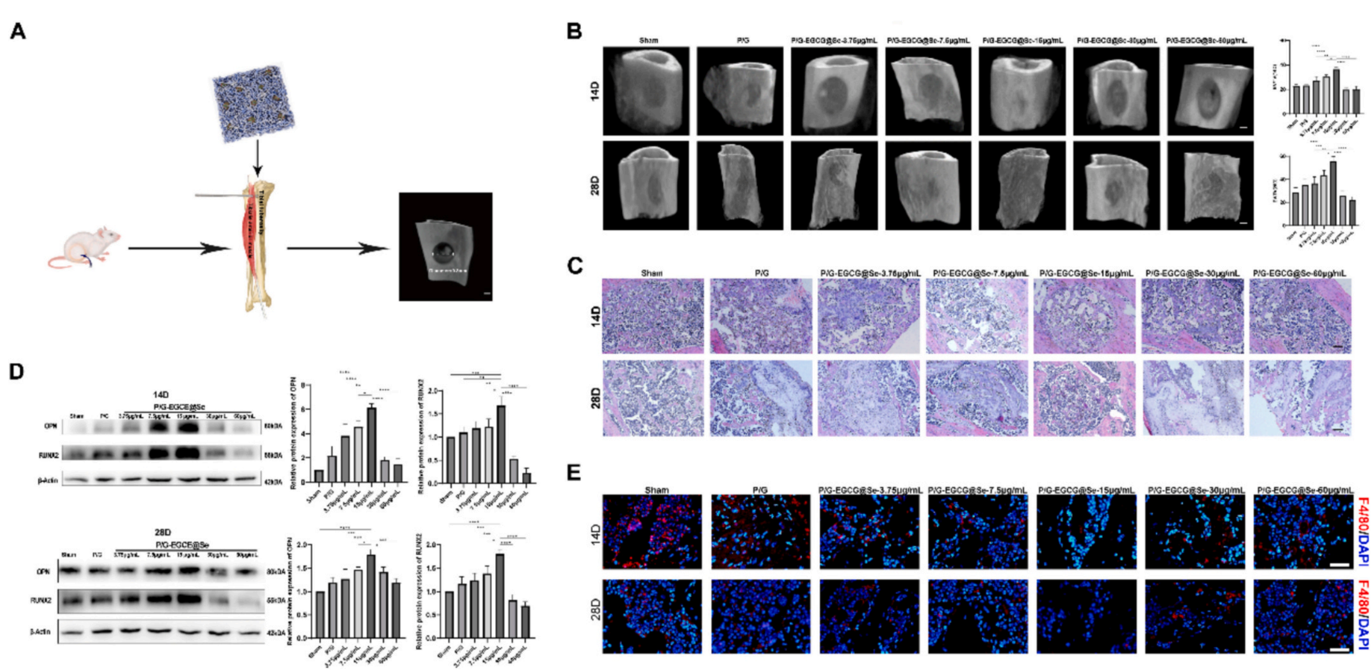


Fig. 4. *In vivo* osteogenic effect of P/G-EGCG@Se composite nanofibers with different selenium content.
A: Schematic diagram of the mouse tibial defect model and material implantation. B: Micro-CT reconstructed images showing bone regeneration (Bar = 500 μm). C: Hematoxylin and eosin staining to visualize newly formed bone callus in the defect area (Bar = 50 μm). D: Western blot analysis of osteogenesis-related protein expression in bone callus tissue. E: Immunofluorescence staining demonstrating F4/80^{+} macrophages in osteotylus tissue. Statistical analysis was performed using one-way ANOVA followed by Tukey's *post hoc* test. Statistical significance is indicated as follows: *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$, ****$p < 0.0001$. All data are presented as mean ± standard deviation ($n = 3$, independent experiments).

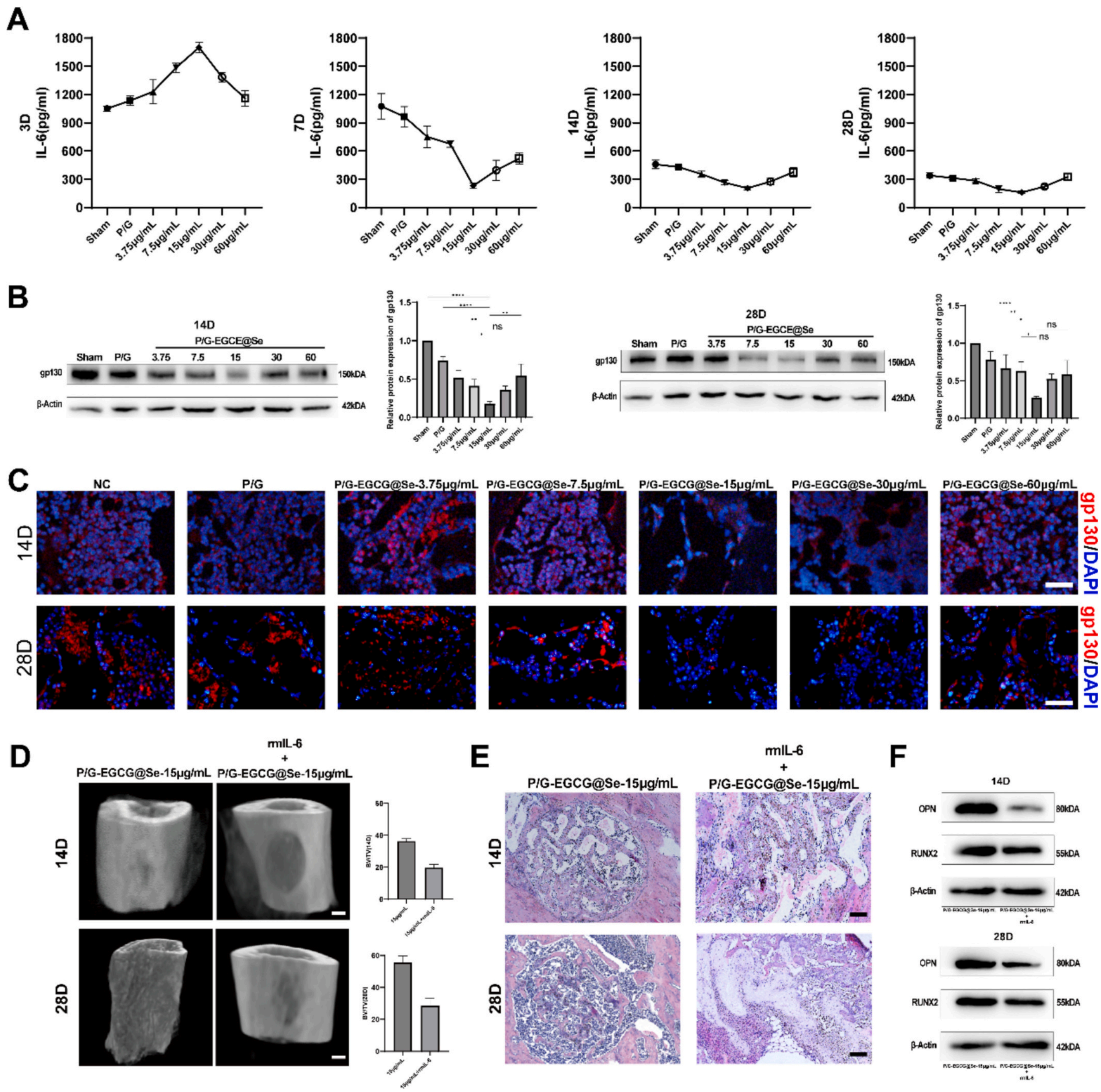


Fig. 5. *In vivo* regulatory effects of P/G-EGCG@Se nanofibers with different Se content on IL-6 and its receptor gp130.
A: ELISA analysis of serum IL-6 levels in mice receiving scaffold implantation at days 3, 7, 14, and 28 post-surgery. B: Western blot analysis of gp130 protein expression in bone callus tissue. C: Immunofluorescence staining for gp130 in bone callus tissue (Bar = 50 μm). D: Representative Micro-CT reconstruction images showing bone regeneration in the defect region of mice implanted with 15 μg/mL Se-loaded nanofibers with or without rmIL-6 intravenous administration (Bar = 500 μm). E: Hematoxylin and eosin (H&E) staining of the defect area in mice implanted with 15 μg/mL Se-loaded nanofibers with or without rmIL-6 administration (Bar = 50 μm). F: Western blot analysis of osteogenic marker proteins in bone callus tissue in mice implanted with 15 μg/mL Se-loaded nanofibers with or without rmIL-6 administration. Statistical analysis was performed using one-way ANOVA followed by Tukey's *post hoc* test. Statistical significance is indicated as follows: $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$, $^{****}p < 0.0001$. All data are presented as mean ± standard deviation ($n = 3$, independent experiments).

(Fig. 5E) and western blot (Fig. 5F) suggested the adding of rmIL-6 blocked bone repair and decreased protein levels of Runx2 and OPN in osteotylus tissue.

To determine the relationship between muscle-derived IL-6 and the observed effects, we conducted an analysis of tibialis anterior (TA) muscle tissues adjacent to the defect, which was damaged by the insertion of a Kirschner wire. Of note, in response to the stimuli of nanofiber implants, the expression levels of the myogenic transcription factor Myogenin and the structural protein Desmin were not influenced by the different Se content on nanofibers (Fig. 6A, B). However, RT-qPCR data indicated that, the implanted nanofiber loading with 15 μg/mL Se significantly reduced IL-6 gene level in the injured TA muscle at seven days post-surgery (Fig. 6C). These results collectively imply that P/G-EGCG@Se nanofibers exhibit osteoinductive properties *in vivo*, and that 15 μg/mL Se loaded-nanofibers effectively inhibits the expression of muscle-derived IL-6, thereby facilitating the repair of bone defects.

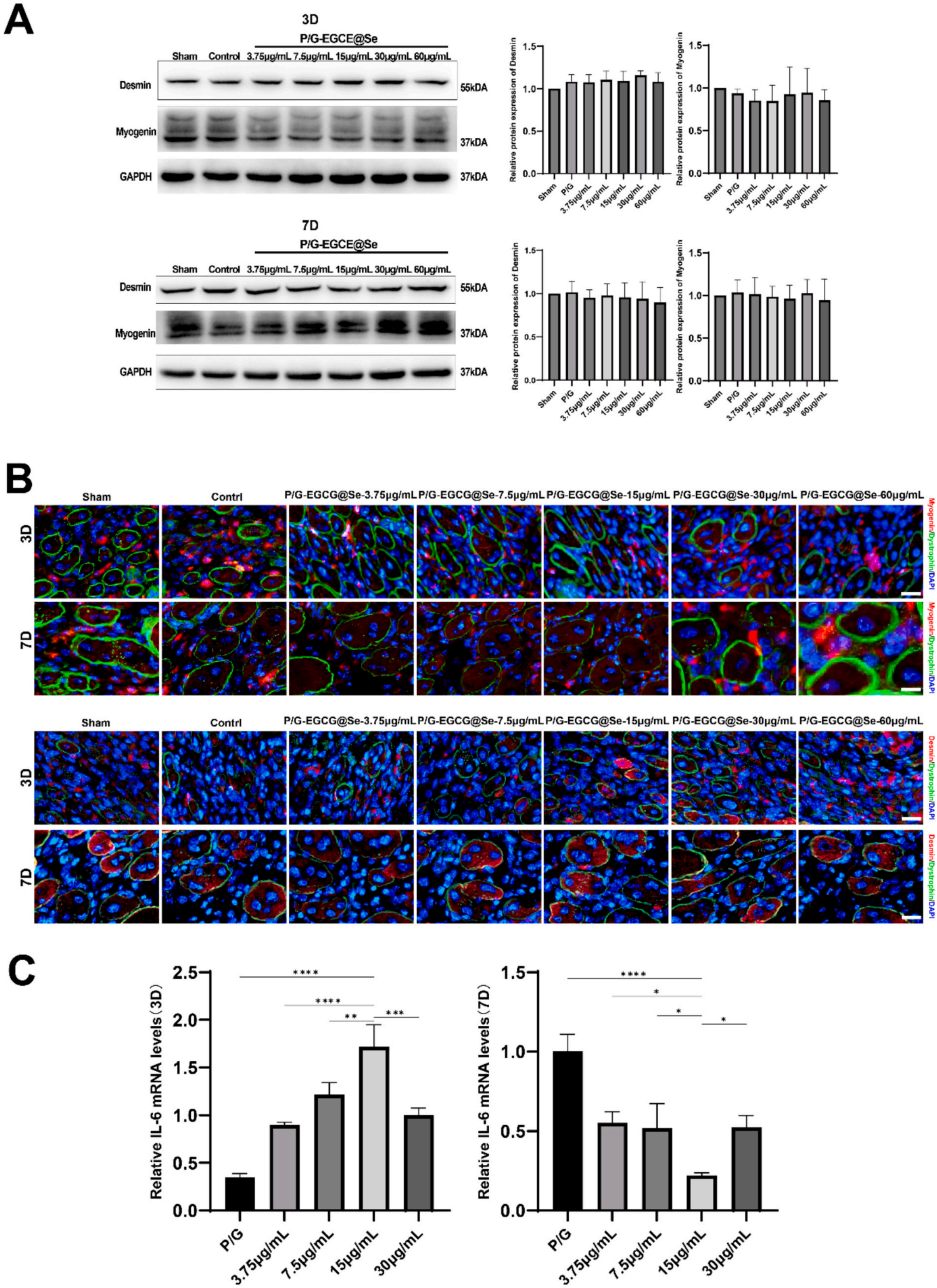


Fig. 6. Implanted P/G-EGCG@Se nanofibers modulate IL-6 synthesis in injured skeletal muscle.
A: Western blot analysis of Myogenin and Desmin protein expression in muscle tissue. B: Immunofluorescence staining of Myogenin and Desmin in muscle tissue (Bar = 50 μm). C: qPCR analysis of IL-6 gene expression in muscle tissue. Statistical analysis was performed using one-way ANOVA followed by Tukey's *post hoc* test. Statistical significance is indicated as follows: *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$, ****$p < 0.0001$. All data are presented as mean ± standard deviation ($n = 3$, independent experiments).

Macrophages dominate a basic inflammatory response in muscle that follows a particular stimulus or injury [31]. Moreover, in inflamed tissue, macrophages is the main source of IL-6 [32]. For these reasons, we next investigated whether nanofibers loading with different Se contents affect on intramuscular infiltration of macrophage. As shown in Fig. 7A, P/G and P/G-EGCG@Se nanofibers induced a markedly F4/80$^+$ macrophages aggregation in inflamed TA muscle than that of sham group on day 3 post-implanting. However, the number of macrophages reduced significantly On day 7. We further monitored that, comparing to sham or P/G implanted mice, there was no statistic difference for absolute numbers of macrophages in inflamed muscle closed to the bone defect implanting with the different Se loaded-nanofibers. Considering that cytotoxic T cells-mediated autoimmune attack of myofibers is the distinct histopathologic feature of myopathies [33], we next detected CD8$^+$ cytotoxic T cells. However, we did not monitor obvious number difference of CD3$^+$CD8$^+$ T cells in inflamed muscle adjacent to bone implants of 3.25, 7.5 or 15 μg/mL Se loaded-nanofibers (Fig. 7B). This indicates that, the nanofibers is relatively biocompatible *in vivo*, which only initiated a light and transient inflammatory response in muscle surrounding the implanting bone. Further, our work suggest Se content on nanofibers had no correlation with muscle inflammation, and plasma IL-6 level decrease was due to P/G-EGCG@Se nanofibers acting on myofibres, but not on macrophages.

In order to investigate the direct impact of Se loaded-fibers on muscle cells, we next turned to *in vitro* experiments. C2C12 myoblasts were cultured and differentiated about 72 h in horse serum into Desmin-positive multinucleated myotubes (Fig. 8A). P/G-EGCG@Se nanofibers were introduced into the culture system and incubated for 6 h. Subsequent PCR and ELISA analyses demonstrated that, 3.75, 7.5, 15 and 30 μg/mL Se loaded-fibers effectively inhibited IL-6 synthesis in muscle cells. Notably, 15 μg/mL Se loaded-nanofibers exhibited the prominent reduction in both IL-6 gene and protein expression in C2C12 myotubes (Fig. 8B, C). Following the 6-h treatment of C2C12 myotubes with Se loaded-fibers, MC3T3-E1 cells were introduced into co-culturing under osteogenic induction for a period of 14 days. In comparison to the untreated MC3T3-E1 cells, the co-culture with C2C12 myotubes and P/G-EGCG@Se nanofibers led to a significant down-regulation of gp130 protein, but up-regulation of osteogenic molecules Runx2 and OPN proteins in MC3T3-E1 cells (Fig. 8D, E). Of note, 15 μg/mL Se-treated nanofiber resulted in the most significant regulation for gp130, Runx2 and OPN in MC3T3-E1 cells (Fig. 8D, E). For co-culture system involving 15 μg/mL Se-treated nanofiber, C2C12 and MC3T3-E1 cells, adding rmIL-6 effectively reversed the expression increase of Runx2 and OPN in MC3T3-E1 cells. However, this suppression for osteogenic molecules was corrected again by further adding IL-6 neutralizing antibody, or IL-6 signaling antaogonist SC144 (Fig. 8F). Combined, our results illustrated that, the bone defect implanting of 15 μg/mL Se loaded P/G-EGCG@Se nanofiber resulted in a remarkable reduction in IL-6 production from myofibers and contributed to correct the IL-6-depressed osteogenesis.

4. Discussion

Electrospun polymeric fibers can be functionalized with bioactive components, including growth factors, functional nanoparticles, and therapeutic agents. Due to their high surface-area-to-volume ratio, adjustable mechanical properties, and structural versatility, these fibers have got extensive applications in the repair of bone and cartilage defects [1–4]. A significant challenge in enhancing the osteoinductive and osteoconductive properties of these scaffolds is the implementation of surface modification strategies that improve cellular affinity and biological activity. Polycaprolactone (PCL) is a widely utilized polymer in bone tissue engineering scaffolds, and electrospun PCL fibers have been shown to facilitate cell alignment and migration [2]. Numerous investigations have concentrated on compositional modifications of PCL fibers to augment their capacity for bone regeneration. For instance, the incorporation of bone morphogenetic protein-2 (BMP-2)-loaded PLGA microspheres into PCL fibers through electrospraying has been demonstrated to promote the osteogenic differentiation of stem cells and enhance *in vivo* bone regeneration in murine models [34]. Cheng et al. developed a functional PCL nanofiber composite capable of releasing Zn^{2+} ions, thereby creating a conducive microenvironment through a combination of physical (nanoscale topography) and chemical (ion release) signals to facilitate osteogenesis [5]. Furthermore, biomimetic coatings composed of calcium phosphate and selenium nanoparticles

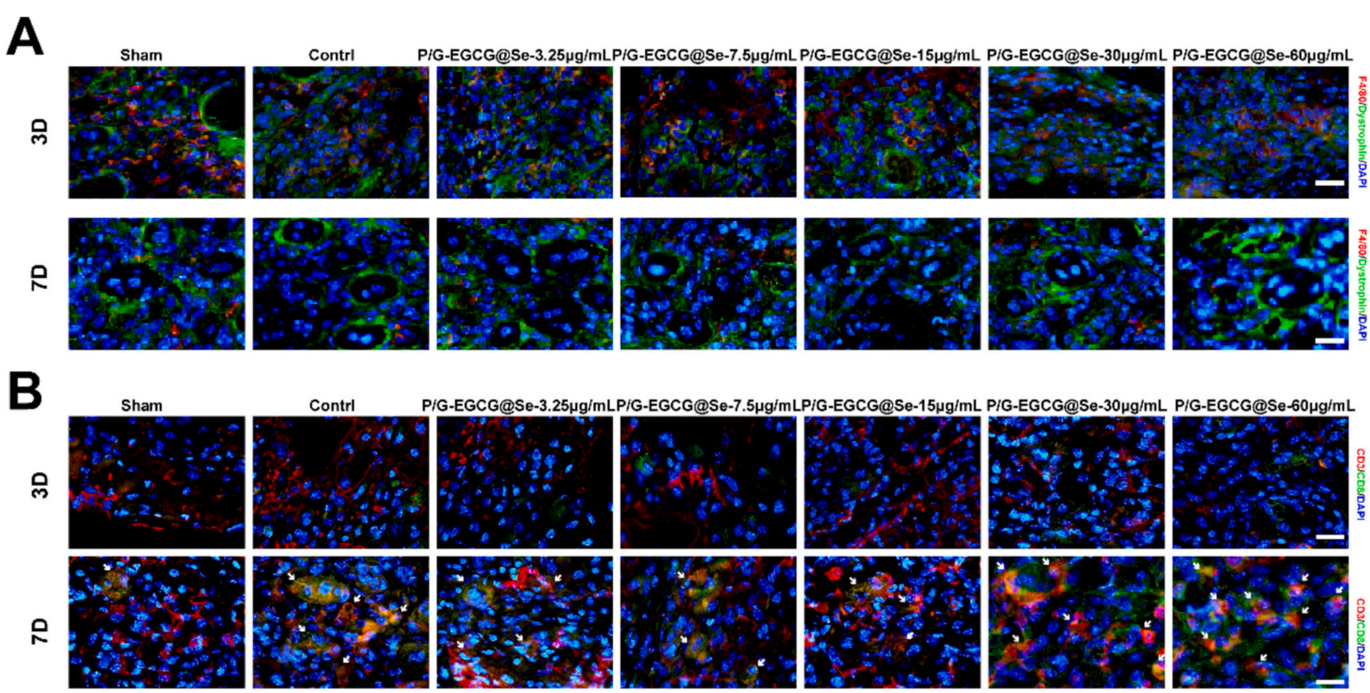


Fig. 7. Immunofluorescence detection of the infiltration of F4/80$^+$ macrophage (A) and CD3$^+$CD8$^+$ cytotoxic T cells (B) in inflamed tibialis anterior (TA) muscle. Bar = 50 μm.

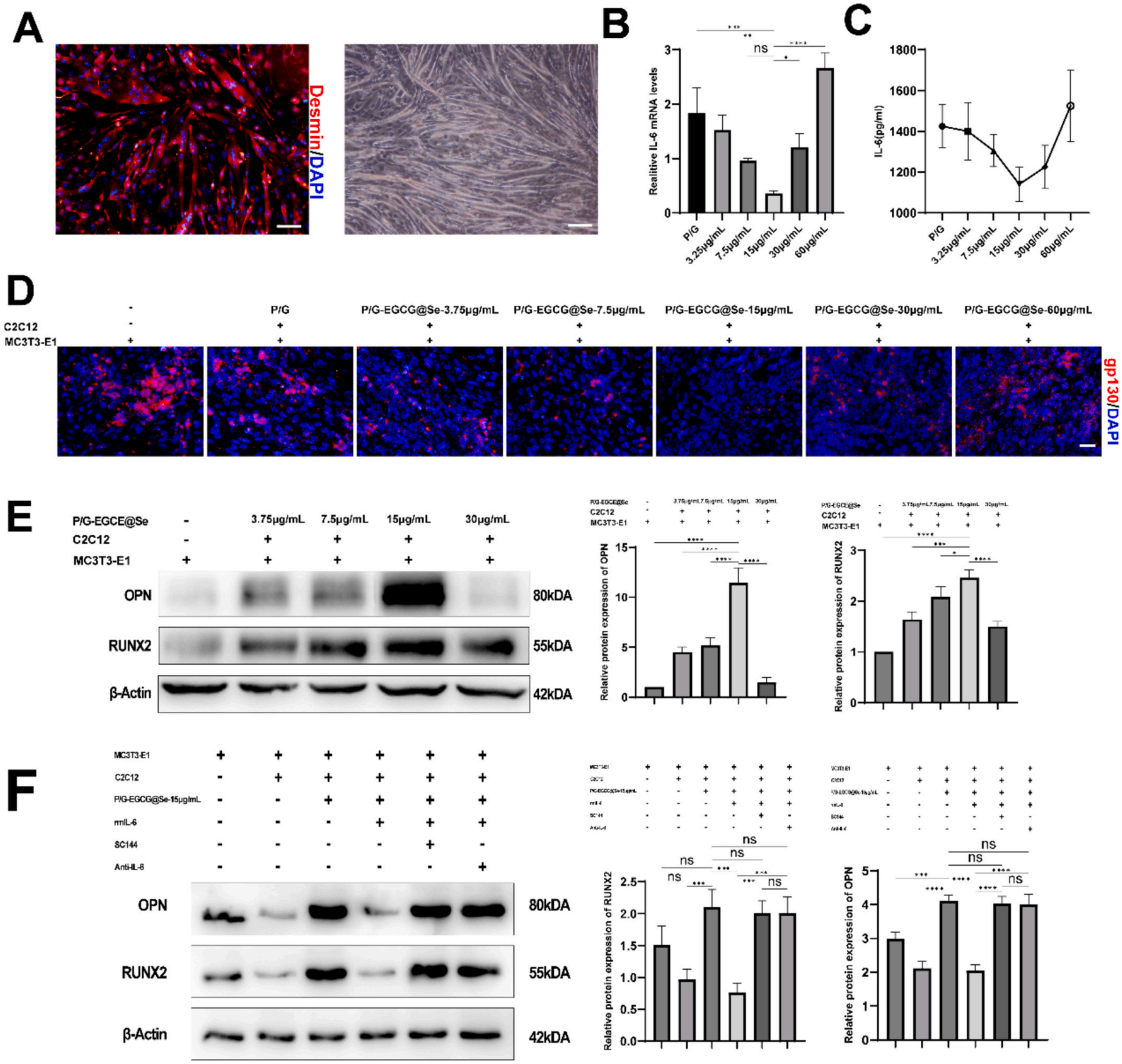


Fig. 8. P/G-EGCG@Se nanofibers with varying Se content regulate osteogenic differentiation *via* modulation of muscle-derived IL-6.
A: Morphological characteristics of C2C12 myotubes after 72 h of differentiation in myogenic induction medium (Bar = 100 μm). B: qPCR analysis of IL-6 gene expression in C2C12 myotubes. C: ELISA measurement of IL-6 protein levels in C2C12 myotube culture supernatants. D: Immunofluorescence staining of gp130 expression in MC3T3-E1 cells co-cultured with C2C12 myotubes (Bar = 50 μm). E: Western blot analysis of Runx2 and OPN protein expression in MC3T3-E1 cells under co-culture conditions. F: Western blot analysis of the effects of Recombinant IL-6 (rmIL-6), IL-6 neutralizing antibody, or IL-6 signaling antaogonist SC144 on the expression of osteogenic markers in MC3T3-E1 cells in the co-culture system. Statistical analysis was performed using one-way ANOVA followed by Tukey's *post hoc* test. Statistical significance is indicated as follows: $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$, $^{****}p < 0.0001$. All data are presented as mean ± standard deviation (n = 3, independent experiments).

applied to 3D-printed PCL scaffolds have exhibited both antibacterial and osteogenic properties [35]. In this work, we fabricated randomly oriented PCL/gelatin (P/G) fibers *via* electrospinning. Our findings indicate that fibers with a PCL-to-gelatin mass ratio of 8:2, produced under a spinning voltage of ≤12 kV, demonstrated optimal structural stability and effectively supported directional cell polarization.

Selenium (Se), an essential trace element within the human body, is integral to various physiological processes, including antioxidant defense, regulation of inflammation and immune response. Furthermore, it is vital for the proliferation and differentiation of osteoblastic cells. Recent studies have identified selenium nanoparticles (SeNPs) as promising nanomaterials that facilitate osteogenic differentiation. [8,16,18,19]. In addition, epigallocatechin gallate (EGCG), a bioactive polyphenolic compound, has demonstrated the ability to promote osteoblast differentiation and the formation of mineralized nodules, thereby playing a significant role in the maintenance of bone health [20–23]. Motivated by the osteoinductive properties of these agents, we developed an inorganic/organic hybrid nanofiber scaffold by applying EGCG@Se onto electrospun PCL/gelatin fibers. *In vitro* experiments demonstrated that the P/G-EGCG@Se nanofibers enhanced the adhesion, proliferation, and differentiation of both mesenchymal stem cells (MSCs) and MC3T3-E1 cells, with the most significant osteogenic effect observed at the fibers treated with 7.5 μg/mL Se solution. Notably, in contrast to the *in vitro* results, nanofibers treated with 15 μg/mL Se solution exhibited the most effective repair of bone defects *in vivo*. By *in vitro* release profile of Se, we found an initial burst Se release, which is favorable for modulating early-stage inflammation, followed by a sustained release that supports long-term tissue repair. These findings suggest that the early and sustained release of Se from the fiber surface is crucial for bone regeneration, and that a specific content of Se may influence the supportive roles of adjacent tissues during the bone healing process.

Both bone and muscle tissues are derived from the mesoderm and demonstrate significant interrelations in their genetic, physiological, and anatomical development. A growing clinical and experimental evidence indicates that, to function as an endocrine organ, skeletal muscle could release a variety of myokines, including insulin-like growth factor-1 (IGF-1), fibroblast growth factor-2 (FGF-2), myostatin, IL-6, osteoglycin (OGN), matrix metalloproteinase-2 (MMP-2), irisin, and secreted protein acidic and rich in cysteine (SPARC), which are crucial for bone growth, development, and regeneration following injury [36,37]. IL-6, in particularly a significant myokine, is abundantly expressed in muscle, and muscle cells produce and release IL-6 in response to exercise and muscle contraction. It is well-known that IL-6 affects bone metabolism, as well as glucose and energy metabolism [38–40]. In our study, mice implanted with P/G-EGCG@Se nanofibers exhibited significantly lower plasma IL-6 levels compared to P/G fibers, with the lowest IL-6 levels monitored in animals receiving nanofibers treated with 15 μg/mL Se suspensions. Noteworthy, the tibialis anterior muscle closed to the bone defect demonstrated a markedly reduction in IL-6 expression following the implantation of 15 μg/mL Se treated fibers. *In vitro* experiments further proved that 15 μg/mL Se loaded fibers inhibited both IL-6 gene and protein expression in differentiated muscle fibers. IL-6 has been reported to mediate bone loss induced by estrogen withdrawal in mice [41]. The *in vivo* studies showed in mice overexpression of IL-6 resulted in increased osteoclastogenesis, through a synergistic effect with tumor necrosis factor-α (TNF-α) to activate NF-κB signaling [42,43]. Furthermore, IL-6 contributes to the MMP-dependent degradation of the bone matrix [44], and anti-IL-6 treatment has been shown to decrease osteoclast numbers and bone resorption parameters in calvarial bone cultured with sera from MDX mice [45]. Collectively, the over-expressed IL-6 hinders bone regeneration. Our results suggest that P/G-EGCG@Se nanofibers enhance bone repair in mice, at least in part, by modulating IL-6 production from skeletal muscle in a Se content-dependent manner.

Low-dose SeNPs have been shown to significantly reduce oxidative stress-induced cellular damage and protect cells from further damage by scavenging free radicals and inhibiting inflammatory responses [46]. Research conducted by Yunjia Wang et al. indicated that low concentrations of SeNPs (5 μg/mL) effectively mitigated cobalt nanoparticle (CoNP)-induced cytotoxicity, oxidative stress and apoptosis in C2C12 cells, and promoted the expression of myogenic markers. Conversely, 20 μg/mL SeNPs were found to be detrimental to the viability and functionality of myofibers [47]. In this work, we did not find Se loaded-nanofibers impact on muscle inflammation procedure and trigger myotoxicity, and on the expression of myogenic markers (Myogenin and Desmin), which suggesting P/G-EGCG@Se nanofibers is relatively biocompatible *in vivo*. It is noteworthy that, myofibers are sensitive to Se released from 15 μg/mL Se loaded nanofibers, which initiates down-regulation of myokine IL-6 production. We had noted that SeNPs were uniformly distributed across the surface of P/G-EGCG@Se fibers. An increase in SeNP loading resulted in a progressively rougher fiber surface and a decrease in the water contact angle, indicating enhanced hydrophilicity. Based on these observations, we hypothesize that, the physicochemical properties of 15 μg/mL Se-loaded P/G-EGCG@Se fibers, involving mechanical properties, surface roughness, the hydrophilicity, are optimal and facilitate cell adhesion, activity, and functional differentiation at the bone defect site. Further, the release of SeNPs from the implants impaired IL-6 synthesis in adjacent muscle tissue, thus corrected IL-6 dependent bone resorption. Previous studies by Chen and Liu have suggested that SeNPs may induce osteogenic differentiation through the BMP/Smad signaling pathway [18,19]. Our earlier work suggests that P/G-EGCG@Se fibers enhance MSCs osteogenesis *via* the activation of the Smad pathway, in which selenium plays a crucial regulatory role (unpublished). Notably, Smad signaling is also a critical regulator in the differentiation of myogenic progenitor cells into mature myofibers [48–50]. In future, we will explore the details of IL-6 signaling pathway in myofibers induced by P/G-EGCG@Se fibers.

Collectively, the data presented here show that, *in vivo*, the special physicochemical properties of P/G-EGCG@Se fibers, assisted by bone repair effects of EGCG, and by the optimal Se content induced IL-6 down-regulation from muscle tissue, contribute to the efficient bone defect healing. Our work thus provides a rationale for designing of ideal bone repair scaffold through interaction among bone and the surrounding non-bone tissues.

CRediT authorship contribution statement

Weichao Gui: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **Yan Deng:** Validation, Software. **Xiaolong Zhang:** Validation, Software, Methodology, Investigation. **Yangyang Li:** Investigation. **Qisen Wang:** Investigation. **Xiaoting Jian:** Investigation. **Jingwen Huang:** Investigation. **Ziwei Zhao:** Investigation. **Qianqian Yu:** Writing – review & editing, Conceptualization. **Hua Liao:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization. **Jijie Hu:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

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Declaration of competing interest

The authors declare no competing interests.

Data availability

Data will be made available on request.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.bioadv.2026.214814.

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